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Short communication



First-principles investigation of optical and thermoelectric properties of $Na_2InCu(F/Cl)_6$ double perovskites for energy harvesting devices

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ABSTRACT

The structural stability, optical, electronic, and thermoelectric properties of $Na_2InCu(F/Cl)_6$ double perovskites have been investigated using density functional theory (DFT) with the Tran–Blaha modified Becke–Johnson potential and spin–orbit coupling (TB-mBJ + SOC). The structural analysis confirms the stability of the cubic phase for both compounds, with lattice parameters of 8.47 Å for Na_2InCuF_6 and 10.07 Å for $Na_2InCuCl_6$. The electronic properties reveal semiconducting behavior with a direct band gap for both materials. Optical parameters, including absorption coefficients, refractive index, and reflectivity, are calculated and analyzed. The band gaps, along with favorable optical properties such as high absorption coefficients, indicate the potential of these compounds for solar cell and optoelectronic applications. Thermoelectric properties, including electrical conductivity, thermal conductivity, and the Seebeck coefficient, are evaluated to determine their contribution to the figure of merit (ZT). The room-temperature ZT values highlight the suitability of $Na_2InCu(F/Cl)_6$ for thermoelectric applications.

1. Introduction

The continuous increase in the global population has significantly heightened the demand for traditional energy resources [1,2]. The over-exploitation of fossil fuels has caused severe environmental pollution and an impending scarcity of energy supplies. To address these challenges, considerable efforts are being directed toward the development of clean and environmentally friendly renewable energy sources [3,4]. Among these, solar energy stands out as a sustainable, safe, and eco-friendly option. Solar energy plays a crucial role in applications such as thermoelectric generators, solar panels, and luminescent devices [5–9]. A critical challenge, however, is identifying suitable materials to advance and optimize solar and renewable energy technologies [10,11]. In this context, perovskite materials have emerged as some of the most promising candidates [12,13].

Perovskites have garnered significant attention due to their remarkable properties, including low cost, ease of synthesis, and high power conversion efficiency [14,15], which make them ideal for optoelectronic devices [16,17]. Beyond photovoltaics, perovskites are utilized in spintronics, fuel cells, memory devices, solid electrolyte batteries, and other applications [18,19]. In particular, double perovskites have gained interest owing to their complex structures, which allow the incorporation of a wider variety of elements into their crystal lattices [20]. This structural complexity can result in unique electronic environments, enhanced environmental stability, and reduced toxicity compared to simple perovskites.

Double perovskites were first identified in the 1950s [21], and their chemical formula, $A_2BB'X_6$, accommodates a broad range of compositions. Here, A typically represents a large cation, such as an alkali or alkaline earth metal, B and B' are intermediate cations, often transition metals, and X is a halide ion. Recent advancements in computational and experimental techniques have highlighted the potential of double perovskites for diverse applications. For instance, Ali et al. [22] investigated the optoelectronic, structural, and thermoelectric properties of $Cs_2/Rb_2InAgBr_6$, demonstrating their suitability for high-performance thermoelectric and photovoltaic devices. Similarly, Ejjabli et al. [23] explored $K_2AgSbBr_6$, a halide double perovskite, and confirmed its potential as a high-efficiency light absorber spanning the ultraviolet and visible spectra. Experimentally, Aguilar et al. [24]synthesized Gd_2MnFeO_6 using the sol–gel method, reporting a narrow optical bandgap of 1.09 eV, along with magnetic properties suitable for magnetically separable catalysts. Furthermore, DFT studies have shed light on the properties of various double perovskites. For example, Khan et al. [25] evaluated Rb_2XCrCl_6 (X = K, Na), which

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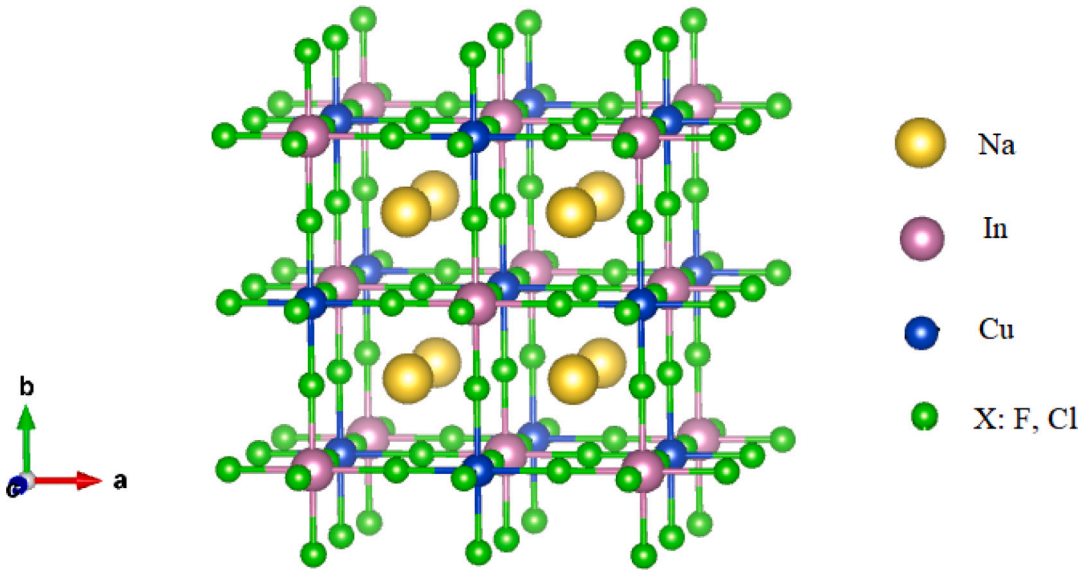


Fig. 1. The crystal structure of $Na_2InCu(F/Cl)_6$ plotted using VESTA software.

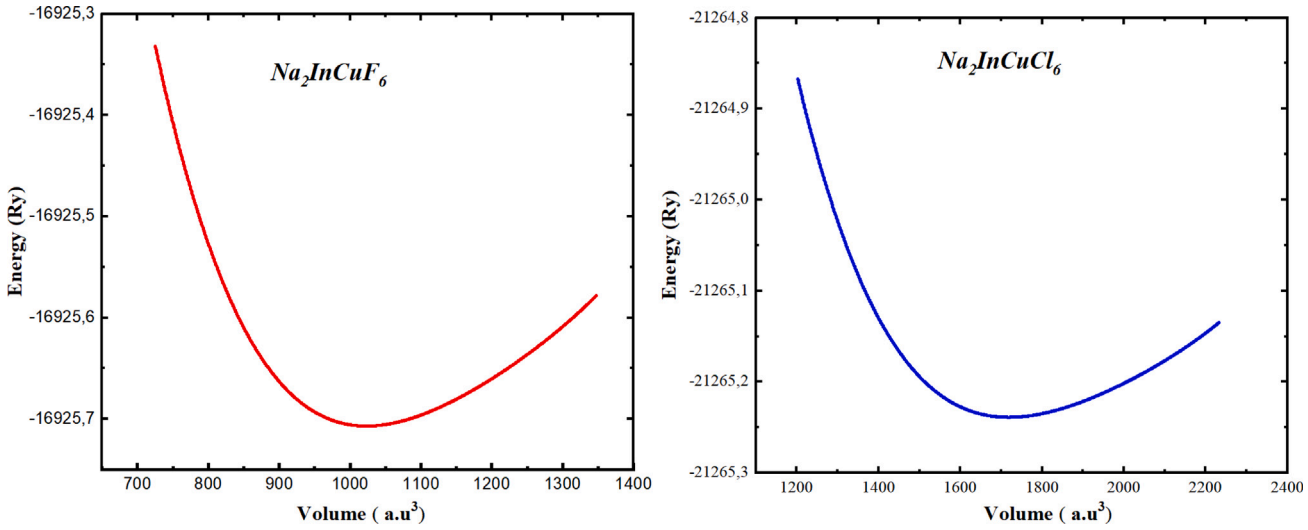


Fig. 2. Graphs illustrating the variation of energy as a function of volume of $Na_2InCu(F/Cl)_6$ double perovskites.

showed semiconductor bandgaps suited for photovoltaic devices and room-temperature detectors. Additionally, Khan et al. [26] studied rare-earth quaternary chalcogenides $BaQCuS_3$ (Q = Er, Gd) and identified promising optoelectronic and thermoelectric properties. These advancements underscore the pivotal role of double perovskites in the development of next-generation renewable energy and electronic technologies.

According to the literature review, there are no existing reports on $Na_2InCu(F/Cl)_6$ double perovskite. However, the latter offers a distinctive advantage over other commonly studied double perovskites, such as those based on bromide or iodide anions. The combination of fluorine and chlorine gives it superior chemical and thermal stability, while enabling fine-tuning of its electronic and magnetic properties, making it promising for applications in optoelectronic and magnetics. Thus, we concentrated this study on examining the crystal structure, electronic, optical, and thermoelectric behavior of $Na_2InCu(F/Cl)_6$ double perovskite, using GGA+TB-mBJ+SOC computations. The organization of this paper is as follows: Section 2 outlines the computational methods used. Section 3 delivers an in-depth analysis of the findings. Section 4 summarizes the conclusions drawn from the study.

2. Computational method

Density functional theory (DFT) is an effective tool for predicting the structures and physical properties of materials, whether they are real or theoretical. Consequently, the investigation on structural stability, optical, electronic and thermoelectric properties of $Na_2InCu(F/Cl)_6$ double perovskites have been explored using DFT based on Tran-Blaha modified Becke-Johnson potential and spin–orbit coupling (TB-mBJ + SOC) [27,28] coded in the Wien2k [29]. In this situation, the wave vector cutoff is set at $R_{mt} \times K_{max} = 8$, with the values of R_{mt} taken as 2.38, 1.75, 1.75, 1.65, and 1.6 atomic units for Na, In, Cu, F, and Cl elements, respectively. In addition, the k-mesh of 20 × 20 × 20 was used for structural electronic properties. The thermoelectric properties of our materials were investigated using the semi-classical Boltzmann transport theory, employing the BoltzTraP code 40 × 40 × 40 [30].

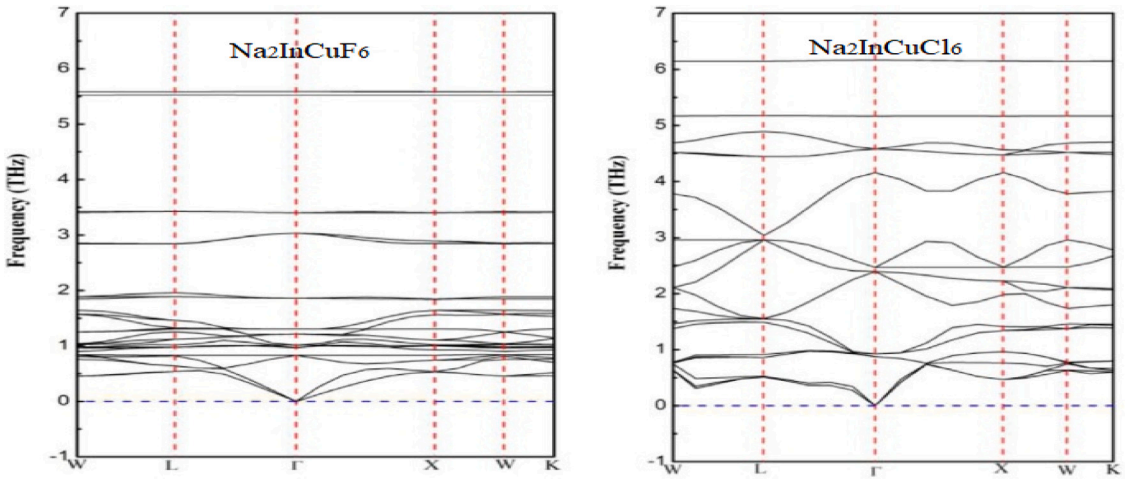


Fig. 3. Dynamic stability of $Na_2InCu(F/Cl)_6$ based on phonon dispersion curves.

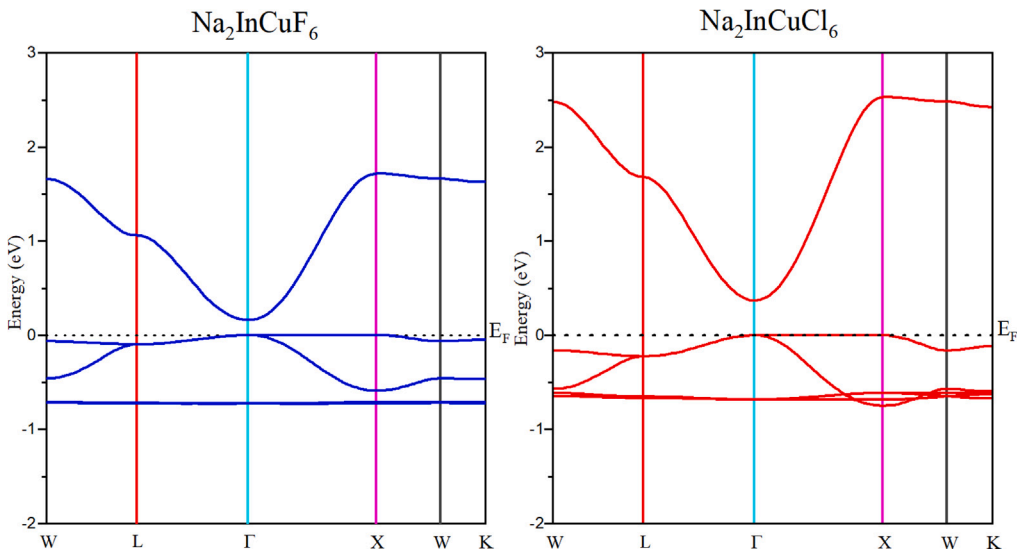


Fig. 4. Band structure plots of Na_2InCuF_6 and $Na_2InCuCl_6$.

3. Results and discussion

3.1. Structure analysis

We begin our results by examining the structural properties of $Na_2InCu(F/Cl)_6$, which share the same cubic perovskite structure with the space group $Fm\bar{3}m$ (#225). Fig. 1 illustrates their crystallographic structure. However, structural stability is of major importance in assessing the physical and chemical features of materials. In order to identify the stable structure of compounds, we optimize the energy as a function of volume curve to obtain the ground state energy and lattice parameter. The Murnaghan equation mentioned in Eq. (1) was used to find the optimal volume and thereby deduce the lattice parameter [31,32].

$$E(V) = E_0 \frac{B_0}{B'(B'-1)} \left\{ V \left(\frac{V_0}{V} \right)^{B'} \right\} + \frac{B_0}{B}(V - V_0) \quad (1)$$

With B and B' are the volume modulus and its derivative, respectively, V_0 is the equilibrium volume and E_0 is the lowest energy at $T = 0$ K.

The results related to the plot of energy as a function of volume are presented in Fig. 2. The computed lattice parameter values are listed in Table 1, with 8.47 Å for Na_2InCuF_6 and 10.07 Å for $Na_2InCuCl_6$. The difference between these values can be attributed to the different ionic radii of the Fe and Cl site ions. Besides, the positions of the different atoms are 8c (0.25, 0.25, 0.25) sites of Na, In - 4b (0.5, 0.5, 0.5) sites, Cu - 4a (0, 0, 0), and F/Cl - 24e (0.25, 0, 0). The Goldschmidt tolerance factor (τ) [33] is a key parameter for ensuring cubic stability in perovskites. It should generally fall within the range of 0.9 to 1.1 to guarantee this stability. In our case, the tolerance factors are 0.96 and 0.94 of Na_2InCuF_6 and $_2InCuCl_6$, respectively, which meet the criteria for cubic stability and thus closely resemble the ideal cubic structure. In order to check the accuracy of the calculated tolerance values,

Table 1
Ground state parameters calculated for Na_2InCuF_6 and $Na_2InCuCl_6$.

Double perovskite	Na_2InCuF_6	$Na_2InCuCl_6$
a_0 (Å)	8.47	10.07
E_0 (eV)	−230 319.5362	−289 328.48
B (GPa)	67.34	36.04
B'	4.82	5.00
τ	0.96	0.94
E_f (eV/atom)	−2.48	−1.32
E_g (eV)	0.17	0.36
ϵ_0	6.25	3.76
n_0	2.50	1.94
R_0	0.18	0.10

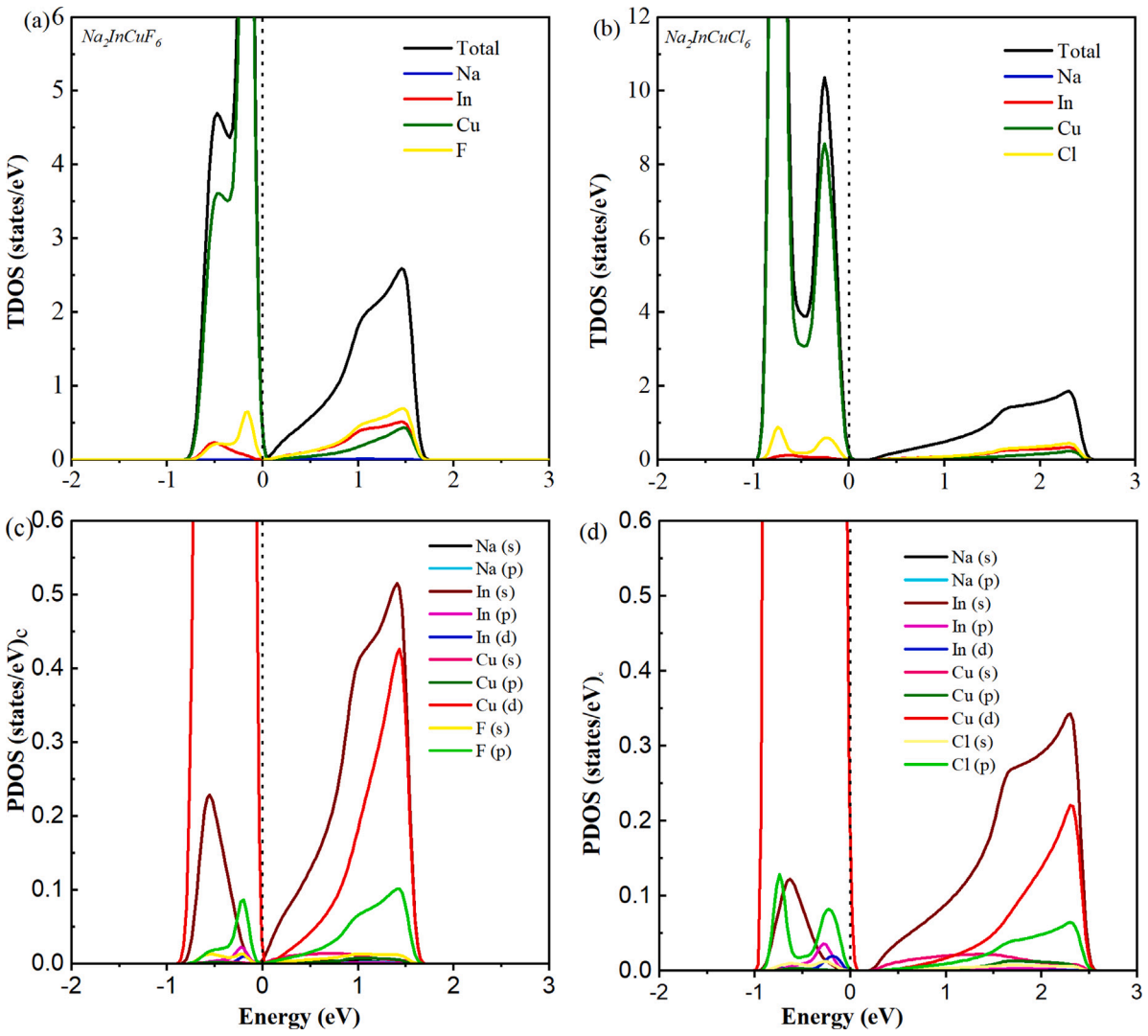


Fig. 5. PDOS and DOS plots of Na_2InCuF_6 and $Na_2InCuCl_6$.

we compare them with previous work on two similar compounds, as reported in Ref. [34].

To appraise the thermodynamic stability of both compounds, the formation energy was determined by employing the equation [35,36]:

$$E_F = E^{\text{Total}}_{\text{Na}_2\text{InCu(F/Cl)}_6} - \left(2E_{\text{Na}} + E_{\text{In}} + E_{\text{Cu}} + 6E_{\text{F/Cl}}\right) \quad (2)$$

$E_{\text{Na}_2\text{InCu(F/Cl)}_6}$ designs the total energy of the studied materials, whereas E_{Na}, E_{In}, E_{Cu} and $E_{\text{F/Cl}}$ represent the energies of the isolated atoms Na, In, Cu, and F/Cl, respectively. As indicated in Table 1, the formation energy for both compounds is negative, reflecting their stability. In other words, it can be synthesized experimentally. The double perovskites $Rb_2AgAsCl_6$ [37] and $K_2AgAsBr_6$ [38] have formation energy values nearly identical to those of Na_2InCuF_6 and $Na_2InCuCl_6$, respectively.

To further validate the structural stability of $Na_2InCu(F/Cl)_6$, phonon dispersion analyses were performed, as illustrated in Fig. 3. Understanding a material's dynamic and thermodynamic behavior requires details on phonon dispersion [39]. These curves are drawn along the high-symmetry directions of the Brillouin zone, namely W-L-Γ-X-W-K. It is clear from this figure that, for both compounds, all phonon modes show positive frequencies throughout the Brillouin zone, reflecting the dynamic stability of the compounds investigated. The same behavior was observed in previous works [40,41].

3.2. Electronic properties

Now, in order to explore the potential application of $Na_2InCu(F/Cl)_6$ in electronic devices, we have determined its electronic band structures, as shown in Fig. 4. The figure clearly shows that the two predicted double perovskites Na_2InCuF_6 and $Na_2InCuCl_6$ possess a direct band gaps equal to 0.17 eV and 0.36 eV, respectively. These results show that both compounds demonstrate a semiconducting behavior [42]. Our materials have a direct bandgap, since in both compounds the conduction band minima and the valence band maxima lie at the Γ-point in the first Brillouin zone. As a result, the modest band gaps reported (0.17 eV to 0.36 eV) do indeed present limitations for some applications, such as photovoltaics, due to significant thermalization losses, which reduce energy conversion efficiency by dissipating excess photon energy as heat [43]. However, these small band gaps can still be advantageous in specific applications, such as infrared detection, thermophotovoltaics or other optoelectronic devices designed for low-energy regimes [44].

On the other hand, the band structure calculations are complemented by the total density of states (DOS) and partial density of states (PDOS) analyses to validate the results. Fig. 5 displays the PDOS and DOS corresponding to the $Na_2InCu(F/Cl)_6$ double perovskites determined using the TB-mBJ + SOC method, ranging from −2 to 3 eV. The vertical dashed line at 0 eV represents the Fermi level. However, we observe that for Na_2InCuF_6, the maximum valence band and the

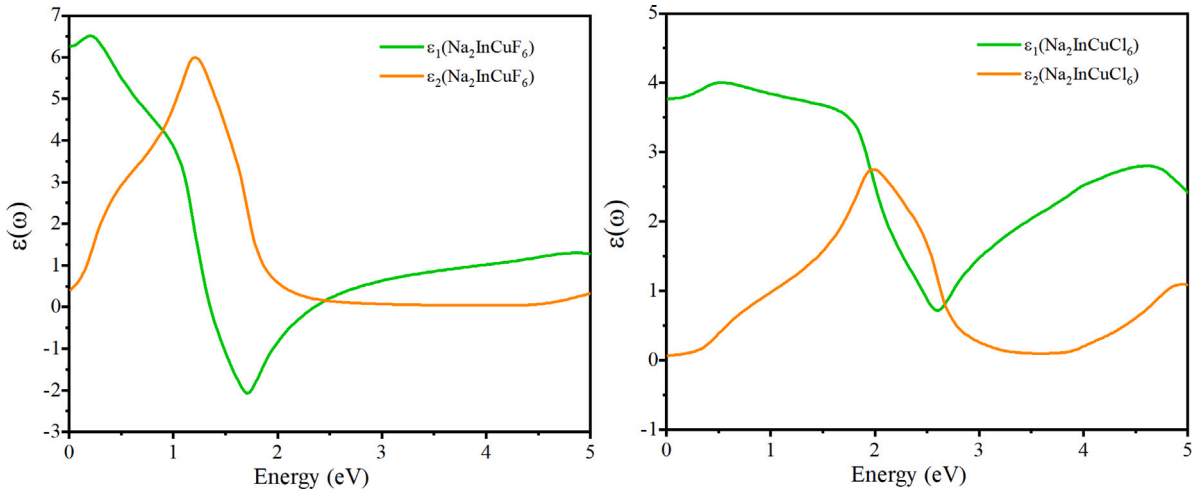


Fig. 6. Dielectric-constants versus energy of $Na_2InCu(F/Cl)_6$ double perovskites.

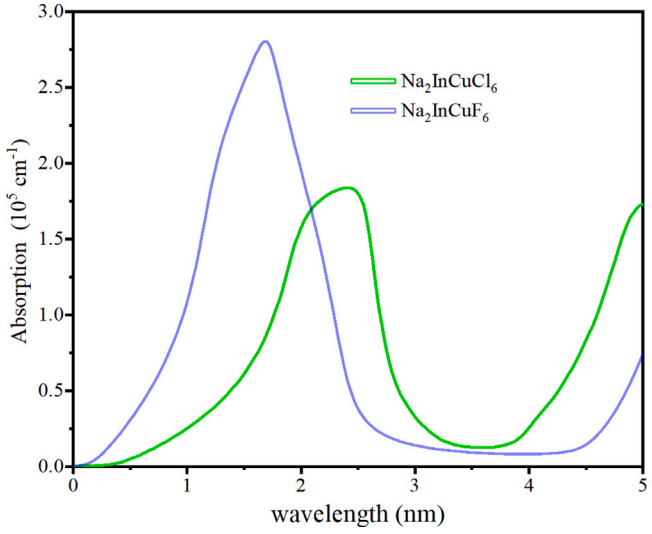


Fig. 7. Absorption spectra versus energy of $Na_2InCu(F/Cl)_6$ double perovskites.

minimum conduction band are dominated by the d orbital of the Cu and In atoms, with a minor contribution from the p orbital of the F atom. This is consistent with the DOS results, which show that Cu has the greatest influence. In contrast, for $Na_2InCuCl_6$, it is evident that in the maximum valence band, the Cu atom with its d orbital plays a predominant role, while the contributions of other atoms, such as In with its d orbital and Cl with its p orbital, are minimal. In the minimum conduction band, the d-orbital of the In and Cu atoms are the main contributors, with a minor contribution from the p-orbital of the Cl atom. This is illustrated by the DOS, which indicates a predominant influence of the Cu atom.

3.3. Optical properties

In this section, we will focus on examining the optical properties of our materials, which gives us an idea of how a material reacts to incident light. Indeed, these properties are determined by specific optical parameters, namely, refractive index, reflectivety, absorption and dielectric constant. These characteristics are expressed in terms of the dielectric function given by Ehrenreich and Cohen's formula [45]:

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \tag{3}$$

with $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ are the real and imaginary of dielectric function. respectively. The first term describes the electronic polarization of incident light and transmits this information and the second one determines the energy loss of light. The expressions of the latter are given by [46–48]

$$\epsilon_1(\omega) = 1 + \frac{2}{\pi} P \int_0^{\infty} \frac{\omega' \epsilon_2(\omega')}{\omega'^2 - \omega^2} d\omega' \tag{4}$$

$$\epsilon_2(\omega) = \frac{4\pi^2 e^2}{\Omega} \lim_{q \to 0} \frac{1}{q^2} \sum_{c,v,i} 2\omega_i \delta(\epsilon_{ci} - \epsilon_{vi} - \omega) \langle u_{ci+eq} | u_{ci} \rangle \langle u_{ci+eq} | u_{ci} \rangle^* \tag{5}$$

Additionally, using $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$, we can investigate other optical parameters such as refractive index $n(\omega)$, absorption coefficient $I(\omega)$ and reflectivity $R(\omega)$.

$$R(\omega) = \frac{n + ik - 1}{n + ik + 1} \tag{6}$$

$$n(\omega) = \left(\frac{\sqrt{\epsilon_1^2(\omega) + \epsilon_2^2(\omega)} + \epsilon_1(\omega)}{2} \right)^{\frac{1}{2}} \tag{7}$$

$$I(\omega) = \sqrt{2}\omega \left(\sqrt{\epsilon_1^2(\omega) + \epsilon_2^2(\omega)} - \epsilon_1(\omega) \right)^{\frac{1}{2}} \tag{8}$$

The curves of the real and imaginary parts of the dielectric function as a function of energy are shown in Fig. 6(a) and (b). Regarding the real values of the dielectric constant, the static values for the two compounds Na_2InCuF_6 and $Na_2InCuCl_6$ are 6.25 and 3.76, respectively. As the energy increases, they reach a peak around 0.3 eV and 0.5 eV. In addition, as the frequency continues to increase, the value of Na_2InCuF_6 decrease and eventually become negative around 1.7 eV, indicating their metallic nature, while $Na_2InCuCl_6$ remain positive. For the imaginary part, it is obvious that with increasing phonon energy, first peaks at 1.5 and 2 eV are reached for Na_2InCuF_6 and $Na_2InCuCl_6$, respectively.

The absorption coefficient curves are illustrated in Fig. 7. It is clear from the figure that the absorption curve follows the trend of $\varepsilon_2(\omega)$. The similarity is primarily attributed to the underlying nature of their absorption characteristics. Inded, from the figure, we can see that there are certain energy points where both materials have a relatively high level of absorption. In the first part of the infrared region of the electromagnetic spectrum, the absorption coefficient of Na_2InCuF_6 is higher than that of $Na_2InCuCl_6$. The absorption curves increase with energy, reaching a maximum around 1.7 eV for Na_2InCuF_6 and 2.7 eV for $Na_2InCuCl_6$ in the visible region of the electromagnetic spectrum, which means that both materials interact strongly with photons in this

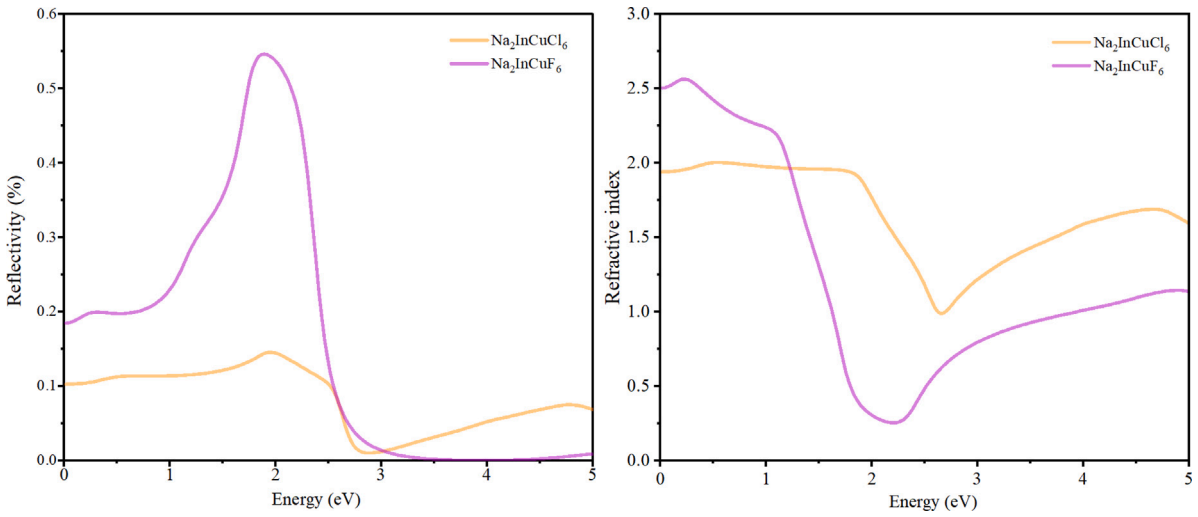


Fig. 8. Refractive-index and reflectivity of $Na_2InCu(F/Cl)_6$ double perovskites.

energy range. We also observe two significant peaks of our materials at around 5 eV with a lower intensity than the maximum peaks, which can be explained by the presence of less likely secondary electronic transitions or interband transitions [49]. Consequently, these components could be highly effective in improving the performance of solar cells, thanks to their ability to absorb light and generate electricity [50].

Fig. 8 displays the refractive-index and reflectivity of $Na_2InCu(F/Cl)_6$. Regarding reflectivity, we can observe that the reflectivity shows almost 2% for Na_2InCuF_6 and 1% for $Na_2InCuCl_6$ at the zero energy point. With increasing energy, the reflectivity increases strongly for Na_2InCuF_6 and weakly for $Na_2InCuCl_6$. For Na_2InCuF_6, the significant peak is observed at 2 eV with 55% reflectivity and for $Na_2InCuCl_6$, it is also observed at 2 eV but with 13% reflectivity. Consequently, Na_2InCuF_6 may be better suited for applications requiring high reflectivity in this wavelength range, while $Na_2InCuCl_6$, with its lower reflectivity, could be more appropriate for applications where low reflectivity is desired, such as in absorbing or anti-reflective materials. An in-depth analysis of these compounds, taking into account their optical properties, reveals remarkable photoelectric properties, illustrating their great potential for use as an absorber layer in a solar cell [51].

Furthermore, any material's refractive index, $n(\omega)$, provides the ratio of the speed of light in a vacuum to the speed of light in that medium. Since $n(\omega)$ provides information similar to $\epsilon_1(\omega)$, both curves exhibit a similar pattern. The computed static refractive index, $n(0)$, for Na_2InCuF_6 and $Na_2InCuCl_6$ was 2.50 and 1.94, respectively. The extracted static values of the real dielectric constant were 6.25 and 3.76, respectively, while the static refractive index values were 2.50 and 1.94 for Na_2InCuX_6 (X = F, Cl), as shown in Fig. 8. The values of both parameters agreed with the relationship $n(0) = \sqrt{\epsilon_1(0)}$ [52–54]. At a visible photon energy of 1.5 eV, the greatest refractive index observed for the $Na_2InCuCl_6$ perovskite was $n(\omega) = 2.2$. When the F atom is replaced by Cl, the maximum refractive index drops to 0.29 at 2.1 eV.

3.4. Thermoelectric properties

Finally, this section focuses on extracting the thermoelectric properties of our compounds $Na_2InCu(F/Cl)_6$. For this reason, the semi-classical Boltzmann theory incorporated in the Boltztrap code is used with the approximation of a constant relaxation time equal 10^{-14}. In general, thermoelectric materials have the ability to convert waste heat into useful electricity during day-to-day operations. They could therefore play a crucial role in energy conservation, and become major resources for tomorrow's energy production. Therefore, the curves of the Seebeck coefficient (S), the electrical conductivity (σ/τ), the thermal conductivity (K/τ) and the figure of merit (ZT) as a function of temperature are shown in Fig. 9(a)–(d). The Seebeck coefficient for the $Na_2InCu(F/Cl)_6$ is illustrated in Fig. 9(a). From this figure, we can see that with increasing temperature from 300 to 900 K, the Seebeck coefficient linearly decreases from 1.8 to 0.9 μV/K for Na_2InCuF_6 and from 1.6 to 0.45 μV/K for $Na_2InCuCl_6$. For the electrical conductivity, as illustrated in Fig. 9(b), the trend observed is the opposite of what is reported in the Ref. [55]; such as with increasing temperature from the same values 300 to 900 K, the electrical conductivity increases linearly from 1.5×10^8 to 7.8×10^8 Ω m s for Na_2InCuF_6 and from 2×10^8 to 7.2×10^8 Ω m s for $Na_2InCuCl_6$. The same behavior was observed for thermal conductivity, except that it increased from 1×10^{14} to 3.25×10^{14} W/mks for Na_2InCuF_6 and from 0.7×10^{14} to 3×10^{14} W/mks for $Na_2InCuCl_6$, as shown in Fig. 9(c). The curves showing the temperature dependence of the figure of merit (ZT) are plotted in Fig. 9(d). As the ZT value reaches its maximum, the material becomes significantly more efficient at converting thermal energy into useful electrical energy [56]. From the figure, an increase in ZT with temperature is evident. For Na_2InCuF_6, ZT increases from 0.5 to 0.53, while for $Na_2InCuCl_6$, it increases from 0.52 to 0.56. These results demonstrate that both materials can maintain the thermal gradient required for efficient thermoelectric energy conversion. The thermoelectric properties of $Na_2InCu(F/Cl)_6$ are consistent with previous findings reported in the literature [57–59]. Consequently, these materials exhibit promising potential for thermoelectric applications. Moreover, a comparison with other materials highlights that hybrid and halide perovskites, such as $SrTiO_3$ [60], $MASnI_3$ [61], $K_2(Se,Te)Br_6$ [62], and Cs_2AgBiX_6 (X = Cl, Br) [63], achieve higher ZT values near unity. Similarly, Cs_2AgSbX_6 (X = Br, Cl) has been analyzed [64], showing tunable direct bandgaps that make them suitable for energy harvesting devices. While $Na_2InCu(F/Cl)_6$ exhibits slightly lower ZT values, its stable thermal gradient and dynamic stability make it a competitive candidate for thermoelectric applications in specific scenarios.

4. Conclusion

In summary, DFT calculations were carried out to investigate the structural, electronic, optical, and thermoelectric properties of $Na_2InCu(F/Cl)_6$ double perovskites using the Tran-Blaha modified Becke-Johnson potential (TB-mBJ) combined with spin–orbit coupling (SOC). The structural parameters at the ground state and the energy versus unit cell volume curve confirm the stability of the cubic phase for both compounds. The electronic properties indicate semiconducting behavior with direct band gaps of 0.17 eV for Na_2InCuF_6 and 0.36 eV

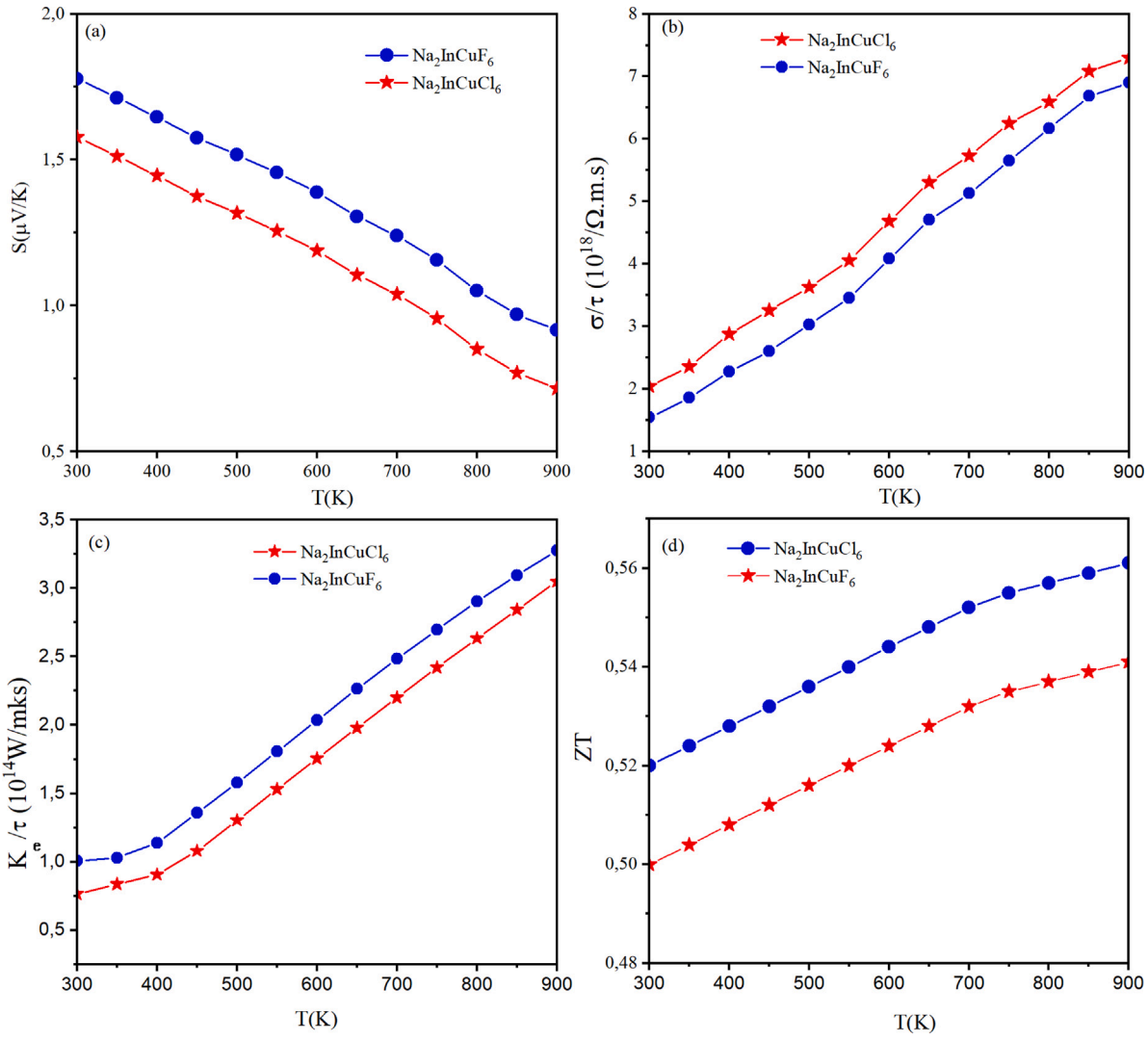


Fig. 9. Thermoelectric coefficients of double perovskites $Na_2InCu(F/Cl)_6$ as a function of temperature. (a) the Seebeck coefficient (S),(b) the electrical conductivity (σ/τ), (c) the thermal conductivity (K/τ) and (d) the figure of merit (ZT).

for $Na_2InCuCl_6$. Optical analysis reveals remarkable properties in the visible region, highlighting their potential for energy technologies such as solar cells. Additionally, the thermoelectric properties demonstrate that both compounds are suitable for thermoelectric applications over a wide temperature range of 300 K to 900 K.

CRediT authorship contribution statement

H. Saadi: Writing – original draft, Software, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **H. Kerrai:** Visualization, Investigation, Formal analysis. **E.M. Jalal:** Software, Investigation, Formal analysis, Data curation. **M. El Bouziani:** Writing – review & editing, Validation, Supervision, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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